

Thesis v2 — Input Coverage Plan

Planning / spec only. No code, no UI, no Deploy work. Repo: [prashanthkrishnan91/claude_financetracker_pk91](#). Branch: [claude/thesis-v2-input-coverage-pNTx5](#).

Decision: Keep suppression / data-quality gates. Do not loosen them to produce better-looking scores. The right path is to improve safe input coverage so quality / growth / risk become publishable when enough real data exists.

1. Context confirmed from repo

- Engine: **v2/backend/app/services/intelligence/thesis_engine.py** — 5 dimensions; MIN_SUBSCORE_QUALITY = 0.40; MIN_CONVICTON_QUALITY = 0.50; INSUFFICIENT_DATA forces WATCH.
- Mapper: **v2/backend/app/services/intelligence/thesis_mapper.py** — yfinance fundamentals + FeatureSet → engine inputs.
- Provider: **v2/backend/app/services/agents/data_sources.py:680–715** *fetch_yfinance_fundamentals_sync* exposes 24 fields from yfinance *info*.
- Orchestrator: **orchestrator.py:1184–1188** calls *map_to_thesis_inputs(raw_funds, fs)* per ticker without asset-class branching.
- Serialiser: **orchestrator.py:62–147** *_scorecard_to_dict* already emits published_dimensions, suppressed_dimensions, missing_dimensions, per-dimension inputs_used / inputs_missing, top-level inputs_used / inputs_missing.
- Reasoning v2: **reasoning_v2_builder.py:526–576** *_build_evidence* already preserves score, published, inputs_used, data_quality, inputs_missing per-dimension under evidence.deterministic.<dim>_score. There is no top-level union.
- History fetch in **data_sources.py:572** uses period="3mo" only — max_drawdown_1y is not derivable today.

Live state confirmed for NVDA / GOOGL: published_dimensions = [valuation, momentum]; suppressed = [quality, growth, risk]; status = INSUFFICIENT_DATA; reasoning_v2 deterministic evidence carries valuation_score and momentum_score; action posture = WATCH.

2. Dimension-by-dimension coverage

A subscore is published when data_quality ≥ 0.40 (≈ ■0.40 × N■ inputs present).

Quality (7 inputs, need ≥ 3 to publish)

Engine input	Status today	Source if added	Verdict
roic_ttm	missing	needs EBIT + tax rate + equity + debt from <i>t.income_stmt</i> / balance sheet — not in <i>info</i>	Future
gross_margin	missing	yfinance <i>info["grossMargins"]</i>	★ Safe — PR B
fcf_margin	available (derived)	already mapped	done
fcf_to_net_income	missing	<i>info["freeCashflow"]</i> / <i>info["netIncomeToCommon"]</i>	★ Safe — PR B
net_debt_to_ebitda	available (derived)	already mapped	done
interest_coverage	missing	needs EBIT + interest expense from <i>t.income_stmt</i>	Future
share_count_delta_3y	missing	<i>t.get_shares_full(start, end)</i> historical	Future

Today: 2/7 (~0.29) → suppressed. Adding **gross_margin** + **fcf_to_net_income** → 4/7 (0.57) → **published**.

Valuation (11 inputs, need ≥ 5 to publish)

Engine input	Status today	Source if added	Verdict
ps_ttm	available	mapped	done
ps_forward	missing	yfinance does not expose forward P/S	Future
p_fcf	missing	market_cap / free_cash_flow (already in fundamentals)	★ Safe — PR B
ev_ebitda	available	mapped	done
peg	available	mapped	done
fcf_yield	missing	free_cash_flow / market_cap (already in fundamentals)	★ Safe — PR B
forward_pe	available	mapped	done
trailing_pe	available	mapped	done
peer_ps_median	missing	needs peer-set + sector dataset	Future / new provider
peer_ev_ebitda_median	missing	needs peer-set + sector dataset	Future / new provider
own_5y_ps_median	missing	needs 5y P/S history series	Future

Today: 5/11 (0.45) → published. Adding **p_fcf** + **fcf_yield** → 7/11 (0.64) — strengthens conviction quality.

Growth (5 inputs, need ≥ 2 to publish)

Engine input	Status today	Source if added	Verdict
revenue_cagr_3y	missing	<i>t.income_stmt</i> historical revenue (3–4y typical)	★ Safe — PR B (extension)
revenue_yoy	available	mapped from revenueGrowth	done
fcf_cagr_3y	missing	<i>t.cash_flow</i> historical free-cash-flow	★ Safe — PR B
gross_profit_yoy	missing	<i>t.income_stmt</i> GrossProfit YoY	★ Safe — PR B
forward_revenue_growth_est	missing	yfinance analyst estimate fields (<i>t.growth_estimates</i>)	Maybe — Future

Today: 1/5 (0.20) → suppressed. Adding any 2 of {revenue_cagr_3y, fcf_cagr_3y, gross_profit_yoy} → ≥ 3/5 (0.60) → **published**.

Risk (7 inputs, need ≥ 3 to publish; gaap_nongaap_gap optional)

Engine input	Status today	Source if added	Verdict
customer_concentration_flag	missing	needs SEC 10-K extraction	Future
guidance_cut_count_4q	missing	needs earnings / guidance data	Future
insider_net_selling_6m	missing	<i>t.insider_transactions</i> + 6m window math	Maybe — PR B follow-up
net_debt_to_ebitda	available (derived)	already mapped	done
beta	available	mapped	done
max_drawdown_1y	missing	extend history window to 1y; deterministic from closes	★ Safe — PR B (provider change)
gaap_nongaap_gap	missing	needs MD&A; / non-GAAP filings parsing	Future

Today: 2/7 (0.29) → suppressed. Adding **max_drawdown_1y** → 3/7 (0.43) → **published**. Adding insider_net_selling_6m later → 4/7 (0.57).

Momentum (5 inputs, need ≥ 2 to publish)

Engine input	Status today	Source	Verdict
relative_strength_vs_spy	available	feature_set.relative_strength_30d	done
trend_regime_score	available (proxy)	mapper_trend_to_regime_score (documented proxy)	done
return_5d	available	feature_set	done
return_30d	available	feature_set	done
sma_20_50_signal	available	derived	done

Today: 5/5 → fully published. **No PR work needed.**

3. Unsafe proxy list — do NOT use

- **profit_margin** → **fcf_margin**. Cash-flow margin and accounting margin diverge sharply for capex- or working-capital-heavy firms. Already explicitly avoided in mapper; keep avoiding.
- **return_on_equity** → **roic_ttm**. ROE inflates with leverage; ROIC neutralises capital structure. Cannot substitute.
- **debt_to_equity** → **net_debt_to_ebitda**. Different denominators, different interpretation; already avoided.
- **dividend_yield** → **fcf_yield**. Dividends are a payout policy decision, not a free-cash-flow measure.
- **earnings_growth** → **forward_revenue_growth_est** or **revenue_yoy**. Earnings ≠ revenue; mixing them corrupts the growth dimension.
- **volatility_30d** → **beta** or **max_drawdown_1y**. All three measure different things (idiosyncratic vol vs market sensitivity vs trough loss). Adding volatility_30d would require an engine input contract change, not a proxy.
- Hardcoded category multipliers, sector defaults, peer medians not actually fetched. Anything fabricated.

4. Asset-type handling

Today *map_to_thesis_inputs* runs identically for stocks, ETFs, and crypto. The orchestrator already knows category (entry["category"], lowercased to "crypto" at orchestrator.py:999).

Asset class	Stock-style dimensions that ARE meaningful	Stock-style dimensions that are NOT applicable
Individual stock	quality, valuation, growth, risk, momentum	n/a
ETF	momentum, partial risk (beta, drawdown)	quality, growth, valuation (per-holding fundamentals do not roll up cleanly via yfinance info)
Crypto	momentum, partial risk (drawdown, volatility)	quality, valuation, growth (no revenue / EBITDA / FCF concept)

Recommendation: do **not** force ETFs / crypto into stock dimensions and accept INSUFFICIENT_DATA noise. Instead, in PR C add asset-class-aware diagnostics so non-stock tickers emit honest **not_applicable** markers per dimension rather than *suppressed_low_quality*.

- **Smallest safe step (PR C-1, diagnostics-only)**: in *_scorecard_to_dict*, when category is ETF / crypto, set unpublished_reasons[<dim>] = "not_applicable_<asset_class>" for the dimensions that do not apply, and include an asset-class-aware user_safe_note. No score-math change.
- **Followup (PR C-2, optional)**: add a parallel ETF / crypto scoring frame (not required to unblock the equity book).

5. Reasoning v2 fix (PR A scope)

Per-dimension `inputs_used` / `inputs_missing` are already preserved at `evidence.deterministic.<dim>_score` (`reasoning_v2_builder.py:551–553`). What is missing for downstream observability:

- A top-level **evidence.deterministic.coverage** block aggregating across published dimensions:
 - ■ **inputs_used**: sorted union across published dimensions.
 - ■ **inputs_missing**: sorted union across published dimensions.
 - ■ **published_dimensions**: list (already implicit in keys, made explicit).
 - ■ **suppressed_dimensions**: list.
- Confirm preservation under PARTIAL and INSUFFICIENT_DATA + ≥ 1 published dimension via a new regression test.

Observability-only. No threshold change. No WATCH behavior change. Reasoning_v2 stays dormant (backend-only).

6. Recommended PR sequence

PR	Scope	Files	SQL	LLM	UI	Deploy
A	Reasoning_v2 deterministic evidence: add top-level coverage block (<code>inputs_used</code> , <code>inputs_missing</code> , <code>published_dimensions</code> , <code>suppressed_dimensions</code> aggregated). Tests for PARTIAL + INSUFFICIENT_DATA + ≥ 1 published.	<code>reasoning_v2_builder.py</code> , <code>tests/test_reasoning_v2_builder.py</code> , <code>HANDOFF.md</code>	No	No	No	No
B	Safe yfinance input coverage: add <code>gross_margin</code> , <code>fcf_to_net_income</code> , <code>p_fcf</code> , <code>fcf_yield</code> , <code>max_drawdown_1y</code> , plus <code>revenue_cagr_3y</code> / <code>gross_profit_yoy</code> / <code>fcf_cagr_3y</code> from <code>t.income_stmt</code> / <code>t.cash_flow</code> . Extend <code>fetch_yfinance_fundamentals_sync</code> and history window (1y for drawdown). Mapper additions are pure derivations — no engine changes, no threshold changes.	<code>data_sources.py</code> , <code>thesis_mapper.py</code> , <code>tests/test_thesis_mapper.py</code> , <code>deterministic_fixture</code> , <code>HANDOFF.md</code>	No	No	No	No
C	Asset-class-aware diagnostics (stocks / ETFs / crypto). Plumb category into mapper or <code>_scorecard_to_dict</code> , mark not-applicable dimensions with <code>not_applicable_<asset_class></code> , asset-class-specific <code>user_safe_note</code> . No engine math change.	<code>orchestrator.py</code> (or new <code>thesis_asset_class.py</code>), <code>tests</code> , <code>HANDOFF.md</code>	No	No	No	No
D	Only after PR A + B + C are live-validated: optional debug inspector endpoint or Deploy-signal consumption of <code>reasoning_v2</code> . Out of scope until coverage is real.	TBD	TBD	TBD	Possibly	Possibly

PR B can be split internally if income-statement extraction is large: **B-1** (info-only fields: `gross_margin`, `fcf_to_net_income`, `p_fcf`, `fcf_yield`) and **B-2** (history-derived: `revenue_cagr_3y`, `fcf_cagr_3y`, `gross_profit_yoy`, `max_drawdown_1y`). B-1 alone unlocks **quality** and strengthens valuation; B-2 unlocks **growth** and **risk**.

PR B does not lower any threshold. Suppression gates remain intact. Engine and `reasoning_v2` are unchanged.

7. First implementation prompt — PR A (Sonnet, Medium usage)

Use after Code Committee approves Medium usage. Block is mobile-safe and self-contained.

Task:

Add a top-level `coverage` block to `reasoning_v2.evidence.deterministic` that aggregates `inputs_used`, `inputs_missing`, `published_dimensions`, and `suppressed_dimensions` across all published thesis dimensions. Per-dimension preservation already exists; this is observability-only aggregation, no behavior change.

State:

- `_thesis_v2` published dimensions live example for NVDA/GOOGL: `["valuation", "momentum"]`; suppressed `["quality", "growth", "risk"]`; status `INSUFFICIENT_DATA`; `reasoning_v2.evidence.deterministic` currently exposes valuation score and momentum score with per-dim `inputs_used/inputs_missing`.
- Top-level `evidence.deterministic.coverage` does not exist today.
- `INSUFFICIENT_DATA` must still force WATCH and must still emit insufficient data blocker.
- Reasoning v2 stays backend-only: no API/UIT exposure.

Primary edit target:

- `v2/backend/app/services/intelligence/reasoning_v2_builder.py` (function `build_evidence`)

Test targets:

- `v2/backend/tests/test_reasoning_v2_builder.py`

Fallback reads only if blocked:

- `v2/backend/app/services/agents/orchestrator.py` (`scorecard_to_dict` for serialized shape reference)
- `v2/backend/app/services/intelligence/score_schema.py`

Discovery budget:

- no find/grep/glob for initial exploration
- read primary + tests first
- read fallback files only if blocked, and state why

Required behavior:

- In `build_evidence` after the per-dimension deterministic dict is built, compute:
 - `published_dimensions`: list of `<dim> score` keys actually present in `det`
 - `suppressed_dimensions`: list of dim names (`"quality", "valuation", "growth", "risk", "momentum"`) whose scorecard subcard is `published=False`
 - `inputs_used`: sorted union across the per-dim `inputs_used` lists in `det`
 - `inputs_missing`: sorted union across the per-dim `inputs_missing` lists in `det`
- Attach as `det["coverage"] = {published_dimensions, suppressed_dimensions, inputs_used, inputs_missing}`. Keep existing per-dim entries unchanged.
- When the scorecard is `None` or has no published thesis dimensions, `det["coverage"]` must be `{published_dimensions: [], suppressed_dimensions: [], inputs_used: [], inputs_missing: []}` (no key omission easier downstream parsing).
- Do not change WATCH/INSUFFICIENT_DATA enforcement. Do not change agreement logic. Do not change data quality logic. Do not change analyst evidence.

Tests to add (no broad refactor):

- `INSUFFICIENT_DATA + valuation/momentum published`: `coverage.published_dimensions == ["momentum_score", "valuation_score"]` (sorted), suppressed contains `quality/growth/risk`, `inputs_used` is the sorted union.
- READY scorecard with all 5 published: coverage published dimensions has all 5; suppressed empty.
- Scorecard absent: coverage block is the empty-shape default.
- Per-dim preservation regression: existing tests must still pass; no behavior change other than the new coverage key.

Constraints:

- No SQL, no migrations, no LLM behavior change, no UI change, no Deploy change.
- Smallest safe patch: no unrelated refactors.
- After opening the PR, stop. Do not propose the next implementation prompt.

Timeout budget:

- If you cannot finish cleanly, stop after a checkpoint. Write: files changed, tests run, current status, remaining steps, whether a PR exists.

Final response format must follow CLAUDE.md.

8. Expected output summary

Supabase SQL expected	No
LLM behavior expected to change	No
Frontend / UI expected to change	No
Deploy expected to change	No

9. Final response format (per CLAUDE.md)

Root cause/plan: Coverage gaps in quality/growth/risk drive INSUFFICIENT_DATA → WATCH for tickers like NVDA/GOOGL. Plan keeps suppression gates and improves safe input coverage. PR A (observability), PR B (yfinance-based safe mappings), PR C (asset-class-aware diagnostics), PR D (deferred).

Files changed: none (planning only).

Tests: none (planning only).

Risks: PR B widens yfinance call footprint (income_stmt, 1y history) — may add latency and breaker exposure; mitigate with small per-call timeout and fixture-based tests. PR C must not change engine math.

Supabase SQL: No

HANDOFF.md edited: No + planning-only spec, no implementation in this turn

README.md edited: No + no public/setup changes